Computer Science Tripos – Part II – Progress Report

Next Generation CHERI Tag Controller

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Description of Progress

So far, I have obtained a trace file and successfully written a decoder in C++. I have also written C++ that parses the contents of the file (which contains the DRAM access trace of a particular workload), and simulates the execution of a given tag controller algorithm running on said workload. This produces an output that can be consumed by ChampSim, an existing C++ cache simulator, which currently provides me with some basic metrics, such as tag cache hit and miss frequencies. I have implemented the hierarchical tag controller algorithm detailed in the paper *Efficient Tagged Memory*, and have run successful simulations with it. This code however is currently untested, and there are many opportunities for errors to have slipped in.

Revisiting the Schedule

Status

I am more than halfway through the core of my project – these tasks remain:

- Gain access to the traces on the CL network
- Select a useful set of analysis/visualisation tools for the simulator, and verify with my supervisor

- Implement these tools
- Implement Arm's Morello controller algorithm
- Test both algorithms and evaluate them using the tools, recording my findings in a formal document.

I have not yet begun work on any of the exetnsions of my project.

Difficulties

One great unforseen difficulty was how long decoding successfully would take. A lack of documentation required me to look through some code of an alumnus who once worked with these files, to engage in a long email exchange with said alumnus, and demanded much trial and error.

Another difficulty was finding the correct output trace format my program should use so that ChampSim can parse them. ChampSim has no documentation regarding its trace formats. Again, this had me looking through (ChampSim's) code.

A further difficulty was that my Michaelmas was relatively heavy with lecture courses. I initially took 5 and dropped one after spending some time on it. I did not originally plan this, and my intention only changed after learning more about this year's courses. Thus, my original workplan did not factor this in, leading to me being further behind than expected.

Adapted Workplan

I am taking 2 courses in Lent and 1 in Easter. This leaves me with more time to catch up.

Gaining access to the traces should be straightforward via communications with CL staff. I will implement the Morello algorithm next. After, I will brainstorm some ideas for metrics that the analysis/visualisation tools could provide, and will implement them. This should be done by 07/02.
I will then test the algorithms, evaluate them with the tools, and record my findings. This should be done by 12/02.
This leaves me 4 weeks behind my original schedule. With a total of 3 weeks of slack period remaining in the original schedule (not counting my 2 week safety-net slack period at the end), and this term's lighter lecture load, I should be able to make up for the remaining 1 week task debt.